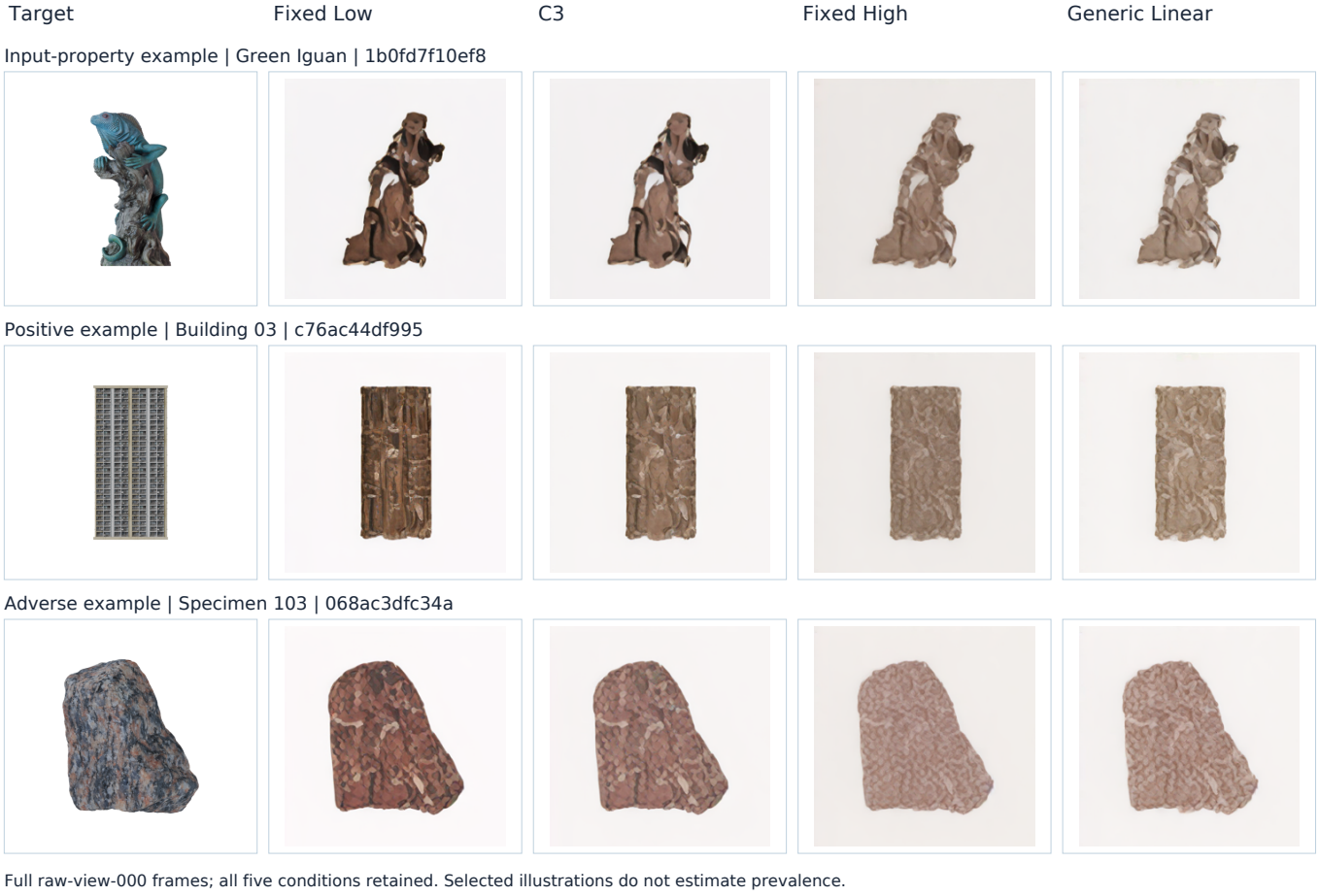


Fig. 2. Fresh C, camera 000. Every row shows Target, Fixed Low, C3, Fixed High and Generic Linear in the same full-frame view. Targets are source RGB conditioning renders, not measured ground-truth albedo. Green Iguan is a colorful, textured organic object selected from source properties. Building 03 (+2.644 dB) and Specimen 103 (−0.468 dB) illustrate positive and adverse C3–Fixed Low FG-PSNR endpoints. These selected rows illustrate endpoint heterogeneity; they do not estimate prevalence or establish a cohort-wide advantage. Complete source identity and credits are in Supplementary Section S9 and the visual selection manifest.

4.3. Appearance and Color Failure Analysis

The color diagnosis is a failure-enriched set of six earlier cases and 20 Fresh C panels, paired on the same inputs, checkpoint and runner across six target views; it is not a random sample, a new holdout or a prevalence estimate. As Table 3 shows, GFL lowered reference-relative CIEDE2000 in 24/26 objects, while pink or purple shifts persisted in two legacy examples. LLH reduced local variation while increasing mean color error, so lower variation did not imply closer appearance. The shifts were present before UV projection or baking; full measurements, VAE/runtime isolates and failed correction candidates are in Supplementary Section S6 and establish no complete cause or repair.

Earlier bake results and calibration checks are reported separately in Supplementary Section S5 and are not pooled with the ten-object cohort. The GeoTex residual branch with an unset scale differs from an explicit Constant 1.0 request under the wrapper caps; we label the latter by its executed setting.

Full-object evidence and its selection. The complete views in Figure 2 and the color failure in Figure 4 use a shared full-frame display for every condition within each object; the existing color diagnostic retains its identical pixel crop. Favorable/adverse C3–GFL examples are retained in Supplementary Section S9. Outcome-selected displays illustrate behavior, not prevalence. The historical texture audit does not preserve a traceable denominator or validated texture-loss labels; we therefore do not infer color-failure prevalence or texture loss from it. Variation statistics alone cannot establish fidelity.

4.4. Actual Baked Appearance: Local Benefit and External Limits

4.4.1. Native-UV protocol and primary result

We selected ten previously unbaked Fresh C objects in UID order using input availability, native UV/material compatibility, triangle count and camera-mask alignment. Only ten of the 300 screened inputs passed these criteria before the cutoff. Their generated-view outputs were known, but their baked outcomes had not been inspected during selection. This compatibility-selected pilot is not a representative sample; five earlier bake objects are reported separately.

Seven conditions share each object's source mesh, reference, seed, six projection cameras, masks, 256-pixel source resolution, atlas resolution, hole filling and eleven unseen evaluation